

סיבוכיות סיבובים של הוכחות אוב אינטראקטיביות באפס ידיעה

The Round Complexity of ZK-IOPs

Tamar Perets

Bar Ilan university, the faculty of engineering | Main chain: Cyber | Facilitator and
Academic Supervisor: Mor Weiss [mor.weiss@biu.ac.il] | Project number: 109 |
22.09.2026

Abstract

Proof systems are systems used to prove things by a prover to a verifier. IOP is a quite new proof system, which is design to combine the IP and PCP probabilistic proof systems. In an IOP, the prover and the verifier has a communication phase in which they interact for number of rounds. Since it was created, many uses for it have been discovered. It is still a quite new object and so there are many research directions on this object. One of those directions is the reductions of the round complexity of an IOP. For example, by general transformation for IOPs that given some IOP outputs new IOP for the same problem with reduced number of rounds.

Another important property of IOP (and proof system in general) is the ZK property, for cases in which we want to save some secret that the prover has from the verifier. The goal of this work is to examine the effect of transformations for rounds reduction on the ZK property. We first looked at the reduction presented at [ABCY22] and proved that it is preserved ZK, but is not always relevant and effective. Then we looked at the reduction represented at [BM88] for IPs. We adjusted it to IOPs and proved correctness and HVZK preservation under reasonable assumption.

Acknowledgment

This project was conducted under the supervision of Dr. Mor Weiss. I would like to thank her for her time, guidance, help, and support throughout the project.

Contents

1	Introduction	2
1.1	Background and Motivation	2
1.2	Preliminaries	2
2	The First Reduction	4
2.1	Reduction Representation	4
2.2	ZK Preservation Proof	5
2.3	The Sumcheck Protocol and The Transformation Disadvantage	7
3	The Second Reduction	7
3.1	Reduction Representation	7
3.2	Correctness Proof	9
3.3	HVZK preservation	14
4	Summery	15

1 Introduction

1.1 Background and Motivation

Proof systems are systems in which there are two sides, one side is a "prover" side, and the other is a "verifier" side. Using a specific system, the prover's goal is to prove something to the verifier (the use at the system will change according to what needs to be proved, the system define the structure of the proof). Proof systems will have the *completeness* property, which will promise that with high probability a honest prover run against honest verifier will succeed in proving something true, and the *soundness* property, which will promise that non-honest prover will not succeed in proving something that is not true, with high probability.

One proof system that is quite new is *IOP* (*Interactive Oracle Proof*) [BSCS16, RRR17]. This system combines the two previous probabilistic proof systems: *IP* (Interactive Proof), in which the prover and the verifier communicate in a few rounds and the verifier can see all the prover's answers, and *PCP* (Probabilistically Checkable Proof), in which the prover outputs only one proof that the verifier has oracle access to it. In an IOP the prover and the verifier communicate, but there might be prover's answers for which the verifier has only oracle access, that is, she should verify the prover proof by reading only a few bits of the prover implicit answers.

Sometimes, the prover has some secret that he wants to hide from the verifier, and then we can define the *ZK* property. That property, intuitively, means that the prover proves his proof successfully without the verifier learning anything (or anything significant, depends on the ZK definition) about the secret except that the statement being proven is correct, even if she is a malicious verifier. This work focuses on IOPs defined on promise problems, and ZK property for promise problems that represent NP languages. In this case the secret will be the witness of some instance is the language.

A main motivation for IOPs is that they enable us to achieve better parameters than the best PCP constructions we have, for example, [BCL21] construct a ZK-IOP that has a linear-time prover and poly-logarithmic-time verifier (within the arithmetic circuit model); such a construction is not known in the PCP model.

Since IOP was created, many uses for it have been discovered. For example - IOPs can be used to construct succinct arguments (which is used in Block Chain) or to prove the hardness of stochastic problems approximation results. Moreover, IOPs can enable us to acquire important properties like fast prover, linear proof length, and more. They also led to a breakthrough in the construction of highly-efficient cryptographic arguments.

IOP is still a new object and there is a lot of research in this field. One of the interests is the round complexity. The goal is to reduce the complexity of the rounds, that is, the number of communication rounds between the verifier and the prover. A state-of-the art work [ARZR25] gives an IOP for large class of boolean circuits, with only $\log^*(S) + O(1)$ rounds when S is the circuit size. As we mentioned, the ZK property is also an important property for cases in which we want to hide some secret that the prover has from the verifier. This property is not easy to preserve, and even tiny changes in the protocol can harm it.

The goal of this work is to focus on the effect of reducing round complexity on the ZK property of the initial IOP. We will focus on IOPs that are both public-coin (i.e., the verifier has no private randomness the prover does not know), and perfectly complete, namely an honest prover succeeds with probability 1.

1.2 Preliminaries

In this section, we will represent some required formal definitions. The next definitions based on [RW24]:

Definition 1 (Interactive Oracle Proof (IOP)). An ℓ -round (public coin) Interactive Oracle Proof (IOP) with soundness error ϵ for a promise problem (YES, NO) is a pair $(\mathcal{P}, \mathcal{V})$ of PPT algorithms, satisfying the following properties.

- **Syntax.**

- **Inputs:** Prover $\mathcal{P}$ receives as input $x = (x_{exp}, x_{imp})$ and a witness w ¹. The verifier $\mathcal{V}$ receives as input x_{exp} and $1^{|x_{imp}|}$, and also has oracle access to x_{imp}

¹If the promise problem does not represent an NP relation, $\mathcal{P}$ does not receive w .

- **Protocol Structure.** The protocol consists of a *Communication Phase* followed by a *Query Phase*. In the *Communication phase*, $\mathcal{P}$ and $\mathcal{V}$ interact for ℓ rounds, where $\mathcal{V}$'s messages are uniformly random strings $R_1, \dots, R_\ell$, and each of $\mathcal{P}$'s messages is either an *explicit* message or an *oracle* message.

In the *Query phase*, $\mathcal{V}$ reads $\mathcal{P}$'s explicit messages in full, and makes (non-adaptive) queries to the implicit input x_{imp} and to $\mathcal{P}$'s oracle messages. These queries are determined solely by the explicit input x_{exp} and $\mathcal{V}$'s randomness strings $R = (R_1, \dots, R_\ell)$. $\mathcal{V}$ then deterministically decides whether to accept or reject based on the answers to these queries, the explicit input x_{exp} , and the randomness string $R_1, \dots, R_\ell$.

- **Semantics.** The protocol satisfies the following semantic properties:

- **Completeness:** If $x \in YES$, then when $\mathcal{V}$ interact with $\mathcal{P}$, it accepts with probability 1.
- **Soundness:** If $x \in NO$, then for any (possibly unbounded) prover strategy $\mathcal{P}^*$, when $\mathcal{V}$ interacts with $\mathcal{P}^*$, it accepts with probability at most ϵ .

Remark. The input is separated into x_{exp} which is the explicit part of the input read by the verifier explicitly, and x_{imp} which is the implicit part to which the verifier has oracle access. In some cases, one of the parts may be empty. (This separation is for cases where we will want verifiers that run at sub-linear time).

Now we will define some more definitions that will help us define ZK-IOP:

Definition 2 (t-Restricted Adversary). We say that an algorithm $\mathcal{A}$ with oracle access to oracles $\mathcal{O}_1, \dots, \mathcal{O}_\ell$ is t-restricted if $\mathcal{A}$ makes at most t queries (in total) to $\mathcal{O}_1, \dots, \mathcal{O}_\ell$.

For the following definitions, we refer to promise problems that represent an NP relation.

Definition 3 (Verifier View). Let $\mathcal{R} = \mathcal{R}(x, w)$ be an NP relation with the corresponding promise problem $(YES_{\mathcal{R}}, NO_{\mathcal{R}})$. Let $(\mathcal{P}, \mathcal{V})$ be an IOP for $(YES_{\mathcal{R}}, NO_{\mathcal{R}})$ and let $\mathcal{V}^*$ be a (possibly malicious) verifier interacting with $\mathcal{P}$ in the protocol. For every $((x_{exp}, x_{imp}), w) \in \mathcal{R}$ the *view* $View_{\mathcal{V}^*}(x_{exp}, x_{imp}, w)$ in the interaction with $\mathcal{P}$ consists of $x_{exp}, 1^{|x_{imp}|}, \mathcal{V}^*$'s random coins, $\mathcal{P}$'s explicit messages, and the answers to $\mathcal{V}^*$'s queries to x_{imp} and $\mathcal{P}$'s oracle messages when $\mathcal{P}$ gets (x_{exp}, x_{imp}, w) as input and $\mathcal{V}$ gets $(x_{exp}, 1^{|x_{imp}|})$ as input and oracle access to x_{imp} . Notice that $View_{\mathcal{V}^*}(x_{exp}, x_{imp}, w)$ is a random variable that depends on the random coins of $\mathcal{P}$ and $\mathcal{V}^*$.

Definition 4 (Adaptive and Non-Adaptive Verifiers). We say that a verifier $\mathcal{V}^*$ with oracle access to $(\mathcal{O}_1, \dots, \mathcal{O}_\ell)$ is non-adaptive if her oracle queries are determined solely by her input and randomness (in particular, without loss of generality, $\mathcal{V}^*$ makes all her oracle queries in a single round). Otherwise, we say that $\mathcal{V}^*$ is adaptive (in particular, each query of an adaptive $\mathcal{V}^*$ might depend on oracle responses to previous queries).

Now we can define ZK-IOP (as defined in [VW26]):

Definition 5 (Perfect, Black-Box, Straight-Line ZK). Let $\mathcal{R}$ be an NP relation with the corresponding promise problem $(YES_{\mathcal{R}}, NO_{\mathcal{R}})$, let $t \in \mathbb{N}$ be a ZK query bound, and let $Q : \mathbb{N} \rightarrow \mathbb{N}$ be a non-decreasing query overhead function. We say that an IOP $\pi = (\mathcal{P}, \mathcal{V})$ for $(YES_{\mathcal{R}}, NO_{\mathcal{R}})$ has perfect black-box straight-line (Q, t) -ZK if there exists a PPT simulator Sim which has black-box straight-line access to the verifier, such that for every t-restricted verifier $\mathcal{V}^*$ and every $((x_{exp}, x_{imp}), w) \in \mathcal{R}$, the following two distributions are identical:

- $REAL_{\pi, \mathcal{V}^*}(x_{exp}, x_{imp}, w) := (View_{\mathcal{V}^*}^{\mathcal{P}}(x_{exp}, x_{imp}, w), Q(q_{\mathcal{V}^*}))_{r_{\mathcal{P}}, r_{\mathcal{V}^*}} : \mathcal{V}^*$'s view when interacting with $\mathcal{P}$ in π when $\mathcal{P}$ gets (x_{exp}, x_{imp}, w) as input and $\mathcal{V}$ gets $(x_{exp}, 1^{|x_{imp}|})$ as input and oracle access to x_{imp} , and where $r_{\mathcal{P}}, r_{\mathcal{V}^*}$ are the random coins of $\mathcal{P}, \mathcal{V}^*$ respectively, and $q_{\mathcal{V}^*}$ is the number of queries that $\mathcal{V}^*$ makes to all her oracles (i.e., x_{imp} and the oracle messages sent by $\mathcal{P}$).
- $IDEAL_{\pi, Sim, \mathcal{V}^*}(x_{exp}, x_{imp}) := (View_{\mathcal{V}^*}^{Sim}(x_{exp}, x_{imp}), q_{Sim})_{r_{\mathcal{V}^*}, r_{Sim}} : \mathcal{V}^*$'s view when interacting with Sim instead of with $\mathcal{P}$ in π with (x_{exp}, x_{imp}) as input (namely, $\mathcal{V}^*$'s oracle queries are answered by Sim), where $r_{\mathcal{V}^*}, r_{Sim}$ are the random coins of $\mathcal{V}^*$ and Sim respectively, and q_{Sim} denotes the number of queries that Sim makes to x_{imp} . Sim 's input in this execution is $x_{exp}, 1^{|x_{imp}|}$, and it has oracle access to x_{imp} .

We say that the IOP has black-box, straight-line t-ZK if it has black-box, straight-line (Q, t) -ZK where Q is the identity function. We say that the IOP has (Q, ∞) -ZK (∞ -ZK, respectively) if it has (Q, t) -ZK (t -ZK, respectively) for any $t \in \mathbb{N}$.

We say that $(\mathcal{P}, \mathcal{V})$ has **Honest-Verifier ZK** (HVZK), if there exists a simulator Sim s.t. for every input, Sim makes at most $Q(q)$ queries to the implicit input (where q is the honest $\mathcal{V}$'s query complexity) and the view of honest verifier $\mathcal{V}$ distribute the same both in the real world (when operates against $\mathcal{P}$) and the ideal world (when operates against Sim). Denote Q -HVZK.

2 The First Reduction

We will start by looking at the reduction presented in [ABCY22]. In this section, we will represent the reduction and prove it is ZK preserving, but then relate to its relevant and useful for some common IOPs.

2.1 Reduction Representation

The reduction goal is to decrease the round complexity, as explained. The idea behind it is to reduce the rounds only to those for which the verifier uses their prover answers. That is, the prover will only send the answers for which the verifier will make an oracle access. Of course, it is impossible to know in advance which prover answers or rounds it will be, so $\mathcal{V}'$ - the reduction verifier - will guess those rounds (this increases the completeness error, but the soundness error stays the same).²

So, the verifier $\mathcal{V}'$ will guess the relevant rounds. Then she will ask for answers and interact with the prover only for those rounds (for $\mathcal{V}$'s random coins - $\mathcal{V}'$ will toss coins herself, randomly, as public coins). Then, at the last stage, she will simulate $\mathcal{V}$ with the random coin toss and $\mathcal{P}'$'s given answers. $\mathcal{V}'$ will return the same answer as $\mathcal{V}$, except for the case that the rounds guesses was incorrect (so $\mathcal{V}$ query at answer $\mathcal{V}'$ does not have) where she reject.

It is important to say that the reduction only works for public-coin IOPs (this is the model we defined here, in which all randomness of the verifier is known to the prover).

The reduction as described in [ABCY22] (Construction 4.4.):

Construction 1. The prover $\mathcal{P}'$ receives as input an instance $\mathbf{x} = (x_{exp}, x_{imp})$ and a witness w , while the verifier $\mathcal{V}'$ receives as input the instance $\mathbf{x} = (x_{exp}, x_{imp})$. They interact as follows.

1. **V'**: Sample a random set of ℓ indices $I := \{i_j\}_{j \in [\ell]} \in \binom{[k]}{\ell}$ with $i_1 < \dots < i_\ell$. For notational convenience, set $i_0 = 0$. Send I to the prover.
2. For $j = 1, \dots, \ell$:
 - (a) **V'**: Randomly sample and send $\rho_{i_{j-1}+1}, \dots, \rho_{i_j} \leftarrow \{0, 1\}^r$.
 - (b) **P'**: Compute and send $\pi_{i_j} := P(\mathbf{x}, w, \rho_1, \dots, \rho_{i_j})$.
3. **V'**: Given direct access to interaction randomness $\rho_1, \dots, \rho_{i_\ell}$ and oracle access to the prover messages $\pi_{i_1}, \dots, \pi_{i_\ell}$:
 - (a) Randomly sample $\rho_{(i_\ell+1)}, \dots, \rho_k \leftarrow \{0, 1\}^r$.
 - (b) Simulate $\mathbf{V}^{\pi_1, \dots, \pi_k}(\mathbf{x}, \rho_1, \dots, \rho_k)$ where if the verifier queries the proof π_j for $j \in I$, the query is forwarded to the appropriate oracle. if **V** queries some $j \notin I$, then immediately reject.
 - (c) Output the same answer as **V**.

At [ABCY22] they prove the following lemma:

Lemma 1. *Let R be a relation with a public-coin k -round round-query IOP $(\mathbf{P}, \mathbf{V})$ with round-query complexity q_{round} . Let ℓ be a parameter such that $q_{round} \leq \ell \leq k$. Then construction 1 yields a public-coin IOP $(\mathbf{P}', \mathbf{V}')$ for R with parameters related to it*

²This effect relevant for more general models that may have a non-zero completeness error. In our model, we have perfect completeness, which is preserved after applying the transformation.

IOP ($\mathbf{P}, \mathbf{V}$) for R	
Completeness error	α
Soundness error	β
Rounds	k
Alphabet size	λ
Proof length (per round)	l
Round queries	q_{round}
Queries (in total)	q
Randomness (per round)	r
Verifier running time	vt

$\longrightarrow$

Round-reduced IOP ($\mathbf{P}', \mathbf{V}'$) for R	
Completeness error	$1 - (1 - \alpha) \cdot \binom{k - q_{\text{round}}}{\ell - q_{\text{round}}} / \binom{k}{\ell} \leq 1 - (1 - \alpha) \cdot (\ell / (e \cdot k))^{q_{\text{round}}}$
Soundness error	β
Rounds	$\ell + 1$
Alphabet size	λ
Proof length (per round)	l
Round queries	q_{round}
Queries (in total)	q
Randomness (per round)	$k \cdot r + \log \binom{k}{\ell} \leq k \cdot r + \ell \cdot \log k$
Verifier running time	$\text{poly}(vt)$

Moreover, if $(\mathbf{P}, \mathbf{V})$ is non-adaptive, then so is $(\mathbf{P}', \mathbf{V}')$

2.2 ZK Preservation Proof

We will now prove that the above reduction preserves ZK. The notation is the same as in the Preliminaries.

Theorem 1. Let $\pi = \mathbf{IOP}(\mathcal{P}, \mathcal{V})$ be a t -ZK IOP for some NP relation $\mathcal{R}$ with the corresponding promise problem $(YES_{\mathcal{R}}, NO_{\mathcal{R}})$ and for some $t \in \mathbb{N}$, and let $\pi' = \mathbf{IOP}(\mathcal{P}', \mathcal{V}')$ be the IOP obtained by applying Construction 1 to $(\mathcal{P}, \mathcal{V})$. Then $(\mathcal{P}', \mathcal{V}')$ is also a t -ZK IOP.

Proof: From the t -ZK property of $(\mathcal{P}, \mathcal{V})$, there exists a PPT *sim* with black-box straight line access to the (maybe malicious) verifier, such that for every $((x_{\text{exp}}, x_{\text{imp}}), w) \in \mathcal{R}$ and every t -restricted verifier $\mathcal{V}^*$ we have:

$$REAL_{\pi, \mathcal{V}^*}(x_{\text{exp}}, x_{\text{imp}}, w) \equiv IDEAL_{\pi, \text{sim}, \mathcal{V}^*}(x_{\text{exp}}, x_{\text{imp}})$$

that is:

$$(View_{\mathcal{V}^*}^{\mathcal{P}}(x_{\text{exp}}, x_{\text{imp}}, w), Q(q_{\mathcal{V}^*}))_{r_{\mathcal{P}}, r_{\mathcal{V}^*}} \equiv (View_{\mathcal{V}^*}^{\text{sim}}(x_{\text{exp}}, x_{\text{imp}}), q_{\text{sim}})_{r_{\mathcal{V}^*}, r_{\text{sim}}} \quad (1)$$

Based on *sim*, We will define the following simulator *sim'* (that runs against some $\mathcal{V}^*$ in $(\mathcal{P}', \mathcal{V}')$. *sim'* uses *sim* as a black-box access.

sim':

Given $x_{\text{exp}}, 1^{|x_{\text{imp}}|}$, an oracle access to x_{imp} and black-box straight line access to $\mathcal{V}^*$:

1. Get $I = \{i_j\}_{j=1}^{j=\ell}$ from $\mathcal{V}^*$. Set $i_0 = 0$.
2. Emulate $\text{sim}(x_{\text{exp}}, 1^{|x_{\text{imp}}|})$ as will be described.
3. For $j = 1, \dots, \ell$:
 - (a) Get $\rho_{i_{j-1}+1}, \dots, \rho_{i_j}$ from $\mathcal{V}^*$.
 - (b) For $k = i_{j-1}+1, \dots, i_j$: Use ρ_k as the message $\mathcal{V}^*$ sent in the protocol. For each k , we denote the k 'th message generated by *sim* as π_k (π_k will be empty if it is an oracle message). if *sim* queries x_{imp} - forward the queries to x_{imp} and return the answer.
 - (c) If π_{i_j} is explicit, send it to $\mathcal{V}^*$. Else (it is implicit), forward each query from $\mathcal{V}^*$ to π_{i_j} to *sim* and return the same answer.

4. Throughout the simulation, if $\mathcal{V}^*$ queries π_{i_j} for some $i_j \in I$, forward it to sim and return the same answer to $\mathcal{V}^*$.

Remark. We can assume that in terms of syntax $\mathcal{V}^*$ behaves like $\mathcal{V}'$ (That is, sending I at the beginning and $\rho_{i_{j-1}+1}, \dots, \rho_{i_j} \in \{0,1\}^r$ for the correct r in every other round) as $\mathcal{P}'$ expects, or otherwise the prover will not cooperate and will terminate.

Note that the environment of sim is as in the real world where he runs against $\mathcal{V}^*$ as in $IOP(\mathcal{P}, \mathcal{V})$. That is because sim' actually acts exactly like $\mathcal{V}^*$ (using $\mathcal{V}^*$).

Time Analysis: sim' only emulates sim (against $\mathcal{V}^*$), and since sim acts as in the real world (where he is interacting with $\mathcal{V}^*$), his runtime will be polynomial, and over all we get that sim' PPT.

ZK Analysis: Let $\mathcal{V}^*$ be a possibly malicious t -restricted verifier that runs against $\mathcal{P}'$, and let $((x_{exp}, x_{imp}), w) \in \mathcal{R}$. We need to show that

$$REAL_{\pi', \mathcal{V}^*}(x_{exp}, x_{imp}, w) \equiv IDEAL_{\pi', sim', \mathcal{V}^*}(x_{exp}, x_{imp})$$

that is:

$$(View_{\mathcal{V}^*}^{\mathcal{P}'}(x_{exp}, x_{imp}, w), Q(q_{\mathcal{V}^*}))_{r_{\mathcal{P}'}, r_{\mathcal{V}^*}} \equiv (View_{\mathcal{V}^*}^{sim'}(x_{exp}, x_{imp}), q_{sim'})_{r_{\mathcal{V}^*}, r_{sim'}}$$

for some $Q : \mathbb{N} \rightarrow \mathbb{N}$, where $r_{\mathcal{P}'}, r_{\mathcal{V}^*}, r_{sim'}$ are the random coins of $\mathcal{P}', \mathcal{V}^*$ and sim' (respectively); $q_{\mathcal{V}^*}$ denotes the number of queries that $\mathcal{V}^*$ makes to all her oracles (i.e., x_{imp} and $\mathcal{P}'$'s oracle messages); and $q_{sim'}$ denotes the number of queries which sim' makes to x_{imp} .

We will define $\mathcal{V}^*$ - a t -restricted verifier that operates in the original IOP $(\mathcal{P}, \mathcal{V})$. $\mathcal{V}^*$ uses $\mathcal{V}^*$ as a black box.

$\mathcal{V}^*$: input: $\mathbb{x} = (x_{exp}, 1^{|x_{imp}|})$ an oracle access to x_{imp} , and a black-box straight line access to $\mathcal{V}^*$.

1. Run $\mathcal{V}^*(\mathbb{x})$. $\mathcal{V}^*$ relays messages from the prover and oracles to $\mathcal{V}^*$ (including access of the oracle to implicit input) and operates as follows.
2. Get $I = \{i_j\}_{j=1}^{j=\ell}$ from $\mathcal{V}^*$. Set $i_0 = 0$.
3. For $j = 1, \dots, \ell$:
 - (a) Get $\rho_{i_{j-1}+1}, \dots, \rho_{i_j}$ from $\mathcal{V}^*$.
 - (b) For $k = i_{j-1} + 1, \dots, i_j$: Send ρ_k to $\mathcal{P}$. Denote the response $\mathcal{P}$ sends as π_k (may be explicit or implicit).
 - (c) If π_{i_j} is explicit, send it to $\mathcal{V}^*$.
4. Continue to emulate $\mathcal{V}^*$, where if $\mathcal{V}^*$ queries some π_i or x_{imp} the query is forwarded to the appropriate oracle.
5. Output the same answer as $\mathcal{V}^*$.

To prove Eq. (1), we will define the following distributions:

- $H_0 = REAL_{\pi, \mathcal{V}^*}(x_{exp}, x_{imp}, w)$
- $H_1 = IDEAL_{\pi, sim, \mathcal{V}^*}(x_{exp}, x_{imp})$

From the ZK of π we have $H_0 \equiv H_1$. Define the following function $f()$, that given H_b ($b \in \{0,1\}$) outputs H'_b that is H_b without the explicit π_k messages of the prover in which $k \notin I$.

Note that $\mathcal{V}^*$ operates with the randomness of $\mathcal{V}^*$ and so she has I in her view, and also that she only queries oracles when $\mathcal{V}^*$ queries. In addition, in the case of H_0 , $\mathcal{V}^*$ bases her answers on $\mathcal{P}$ (similarly to $\mathcal{P}'$), and in the case of H_1 on sim (similarly to sim'). We get the following:

- $H'_0 = REAL_{\pi', \mathcal{V}^*}(x_{exp}, x_{imp}, w)$
- $H'_1 = IDEAL_{\pi', sim', \mathcal{V}^*}(x_{exp}, x_{imp})$

Overall, we get:

$$H_0 \equiv H_1 \Rightarrow f(H_0) \equiv f(H_1) \Rightarrow H'_0 \equiv H'_1 \Rightarrow REAL_{\pi', \mathcal{V}^*}(x_{exp}, x_{imp}, w) \equiv IDEAL_{\pi', sim', \mathcal{V}^*}(x_{exp}, x_{imp})$$

■

2.3 The Sumcheck Protocol and The Transformation Disadvantage

We will now represent a very useful protocol and show why in some cases (like this protocol) this transformation is not good enough. The sumcheck protocol is a very common interactive protocol between a prover $\mathcal{P}$ and a verifier $\mathcal{V}$. We are given some scalar α , $q()$ (a d -variable lower-degree polynomial over appropriate $\mathbb{F}$), and the truth table of $q()$. $\mathcal{P}$ has the truth table and $\mathcal{V}$ has an oracle access to it. The goal is to prove that $q()$ over some hypercube (which contains possible inputs) is summed to α .

The *sumcheck* protocol has $d - 1$ rounds. In round i the prover sends to the verifier the polynomial

$$g_i(X_i) = \sum_{(x_{i+1}, \dots, x_d) \in \{0,1\}^{d-i}} q(r_1, \dots, r_{i-1}, X_i, x_{i+1}, \dots, x_d)$$

when $r_1, \dots, r_{i-1}$ are random values $\mathcal{V}$ sent in previous rounds. $\mathcal{V}$ checks $g_i(0) + g_i(1) = g_{i-1}(r_{i-1})$ and sends a new random value r_i .

As we can see, in the sumcheck protocol, the verifier is using the prover's answer in *each round*. In this case, and since $q_{round} \leq \ell \leq k$ we get $\ell = k$ and in that case the reduction is not reducing the number of rounds (This makes sense, because if the verifier needs every round, we cannot choose only few rounds that will be answered by the prover). So, the introduced reduction is good for some cases (in which the verifier does not use at most of the prover's answers), but for other cases (and common cases) this solution is not enough.

3 The Second Reduction

The first construction we looked at is not enough, and we will have to look at more options. The next reduction we will examine is the one shown in [BM88]. This reduction was originally intended for IPs. In this section we will represent this reduction adapted to IOPs, prove the adapted reduction correctness and relate to it HVZK preservation.

3.1 Reduction Representation

As said, at [BM88] they introduce a round reducing transformation for IPs (which are private cases of IOPs in which the verifier reads all the answers of the prover). We will now describe this reduction, generalized for the IOP model. Given IOP $(\mathcal{P}, \mathcal{V})$, the following parameters are defined:

- t : The number of moves (twice the number of rounds, since every round includes two moves).
- s : The maximum length (bits) of a Prover's possible message in a single round (referring to a full answer, and not only the bits read by the verifier).
- m : $m = 4st$.
- n : the input size.

For this reduction, we assume the following assumptions about the parameters:

1. t is divisible by 4.
2. The given IOP soundness error is $\epsilon \leq (24t)^{-t}$.
3. $t2^{-s-2} + \frac{1}{2t} < \frac{1}{12}$

These assumptions are without loss of generality. Indeed, for assumption (1), if t is not divisible by 4, we can just add meaningless moves so it will. For assumption (2), if the soundness error is too large, then it can be reduced by parallel repetition. This preserves ZK by the composition theorems of [VW26, Thm. 9.1] as long as the initial IOP is perfectly complete (since it is also a public coin). For assumption (3), see remark 3.2 after soundness proof. t, s are defined for the adapted IOP.

What the reduction does, in simple words, is run pairs of close rounds in a single round. The first round runs as usual, but when the verifier sends her randomness, she also sends a few options of randomness for the second round. The prover answers to the first round and to every randomness

option of the second round. Finally, the verifier chooses the option (of second round) to continue with, and they are moving on to the next pair.

Here is the transformation (adapted to the general IOPs case):

Construction 2. Given public coin IOP $(\mathcal{P}, \mathcal{V})$ with soundness error $\epsilon \leq (24t)^{-t}$, let t be the number of moves (divide by 4) in $(\mathcal{P}, \mathcal{V})$, s be the maximal length of a prover's message, and $m = 4st$. The new IOP $(\mathcal{P}', \mathcal{V}')$ is defined as follow.

$\mathcal{V}'$: Input: $x_{exp}, 1^{|x_{imp}|}$ and oracle given access to x_{imp} .

1. Activate $\mathcal{V}(x_{exp}, 1^{|x_{imp}|})$.

Communication Phase

2. For $r = 1, \dots, \frac{t}{4}$:
 - (a) Sample random strings y_0^r (at the length and format of the string $\mathcal{V}$ sends to $\mathcal{P}$ in next round), and $\{y_1^r, \dots, y_m^r\}$ (each at the length and format of the string $\mathcal{V}$ sends to $\mathcal{P}$ in the round after the next) and send them with i_{r-1} (will be defined later) to $\mathcal{P}'$.
 - * At the first round there is no i_{r-1} to send.³
 - (b) Wait for receiving $z_0^r, \{z_1^r, \dots, z_m^r\}$ from $\mathcal{P}'$.
 - (c) Emulate the next round of $\mathcal{V}$. For the randomness of $\mathcal{V}$ use y_0^r , for the answer of $\mathcal{P}$ use z_0^r .
 - (d) Sample $i_r \leftarrow \{1, \dots, m\}$.
 - (e) Emulate the next round of $\mathcal{V}$. For the randomness of $\mathcal{V}$ use $y_{i_r}^r$, for the answer of $\mathcal{P}$ use $z_{i_r}^r$.
3. Send $i_{\frac{t}{4}}$ to $\mathcal{P}'$.
4. If $\mathcal{V}$ sample more randomness at this stage, sample it with the same distribution and sent it to $\mathcal{P}'$ as $y_0^{\frac{t}{4}+1}$, else sent $y_0^{\frac{t}{4}+1} = \perp$.

Query Phase

5. Continue to emulate $\mathcal{V}$, where if $\mathcal{V}$ queries the r 'th prover message for some r or x_{imp} the query is forwarded to the appropriate oracle ($z_{i_r}^r$ or x_{imp}) and the answer is send back to $\mathcal{V}$.
6. Return the same output as $\mathcal{V}$.

$\mathcal{P}'$: Input: $x_{exp}, 1^{|x_{imp}|}$, a witness w , and oracle access to x_{imp} .

1. Active $\mathcal{P}(x_{exp}, 1^{|x_{imp}|}, w)$.

Communication Phase

2. For $r = 1, \dots, \frac{t}{4}$:
 - (a) Wait for receiving $y_0^r, \{y_1^r, \dots, y_m^r\}, i_{r-1}$ from $\mathcal{V}'$.
 - * At the first round there is no i_{r-1} that received.
 - (b) Set the prover message in the previous round to $z_{i_{r-1}}^r$, and discard all other z_i 's.
 - * In the first round- skip this stage.
 - (c) Emulate the next round of $\mathcal{P}$. For the randomness receiving from $\mathcal{V}$ use y_0^r . Denote the message $\mathcal{P}$ want to send back as z_0^r .
 - (d) Emulate the next round of $\mathcal{P}$ for m times, where in the i 'th emulation, use y_i^r as the randomness receiving from $\mathcal{V}$. Denote the message $\mathcal{P}$ want to send back while using y_i^r as z_i^r . (The prover is rewind between emulations to its state in Step 2(c).
 - (e) Send $z_0^r, \{z_1^r, \dots, z_m^r\}$ to $\mathcal{V}'$.
3. Wait for receiving $i_{\frac{t}{4}}, y_0^{\frac{t}{4}+1}$ from $\mathcal{V}'$.

³To be consistent with the original paper, the verifier sends the first message here, meaning she will have one more message than the prover. However, this can be easily adapted to our model by having the prover send an initial empty message.

3.2 Correctness Proof

Theorem 2. Let $\pi = (\mathcal{P}, \mathcal{V})$ be a public coin IOP for the promise problem (YES, NO) with soundness error $\epsilon \leq (24t)^{-t}$, completeness error ψ , t moves (we assume that t is divisible by 4), and maximum prover message length s such that $t2^{-s-2} + \frac{1}{2t} < \frac{1}{12}$ and let $\pi' = (\mathcal{P}', \mathcal{V}')$ be the result of applying construction 2 on π . Then π' is an IOP with a soundness error $\frac{1}{3}$ and a completeness error ψ .

Theorem 2 extends the original proof from [BM88] from the IP to the IOP setting. Moving from the IP to the IOP setting induces some important changes in the proof (compared to the proof given in [BM88]), Specifically, one needs to also consider the query phase and add it to the tree game as well as to adapt the analysis to the new model (where the verifier does not fully read prover messages).

Proof of Theorem 2:

Time: $\mathcal{V}'$ runs $\mathcal{V}$ and $\mathcal{P}'$ runs $\mathcal{P}$ for $\frac{t}{4}$ iterations, $m+1$ times at each iteration. t is the number of moves in $(\mathcal{P}, \mathcal{V})$ and s is the maximal length of a single prover message. Since $\mathcal{V}, \mathcal{P}$ are PPT, t and s are both $\text{poly}(n)$. So, $m = 4st$ is also $\text{poly}(n)$, So they both run $\text{poly}(n)$ algorithms, for $\frac{t}{4} = \text{poly}(n)$ iterations, $m = \text{poly}(n)$ times for each iteration, and $\mathcal{V}', \mathcal{P}'$ are PPT.

Completeness: Let $(\mathbf{x} = (x_{exp}, x_{imp}), \mathbf{w}) \in YES$. When the prover is honest, then the acceptance probability is the same as in the original IOP. Indeed, for every possible choice of $i_1, \dots, i_{\frac{t}{4}}$, the transcript conditioned on $i_1, \dots, i_{\frac{t}{4}}$ is distributed identically to the execution of $(\mathcal{P}, \mathcal{V})$, and so accepted with probability ψ . Using the law of total probability, the verifier accepts with probability ψ .

Soundness: Let $\mathcal{P}'^*$ be a (maybe malicious) prover, and let $\mathbf{x} \in NO$. We need to prove that $\Pr[(\mathcal{P}'^*(\mathbf{x}), \mathcal{V}(\mathbf{x})) = \text{accept}] \leq \frac{1}{3}$.

Since this proof is a little complicated, we will first overview the proof. To prove soundness, we will look at the acceptance chances during the execution, which are the chances that the execution will end with $\mathcal{V}'$'s acceptance (given the partial transcript we look at). The acceptance chances are in relation to π in a similar situation (for situations we can compare them). The main idea is to show that if during π' 's execution for every two merged rounds (that is, one round in π') the acceptance chances are not being improved too much the verifier rejects, and then bound the probability of such improvement during the execution. The proof's steps:

1. Define an IOP tree (which describes the possibilities of IOP execution) and bound the acceptance chances at every stage of possible π 's execution. We also define here projectable nodes in the π' 's IOP tree (which are points in the execution that we can compare π' to π in terms of acceptance chances).
2. Bound the probability of choosing y^r that will allow $\mathcal{P}'$ to improve his acceptance chances too much. That is, y^r for which $\exists z_0^r : |\{i | \exists z_i^r : f(\pi(h), z_0^r, y_i^r, z_i^r) > 12t\delta\}| > \frac{m}{2t}$, see lemma 2.
3. Define a lucky nodes that represent a too large improvement of the acceptance chances and bound their fraction.
4. Prove that if there are no lucky nodes at the full transcript - the verifier reject.
5. Bounding the probability of the existence of a lucky node in a full transcript with $\frac{1}{3}$.
6. Wrapping things up.

Step 1: Define IOP tree and projectable nodes and bound acceptance changes in π

Given an IOP system, we define the IOP tree corresponding to it, that describes all possible transcripts of the IOP execution in the following way: every node represents a (maybe partial) transcript of the IOP, and every child of the node represents the same transcript extended with one possible next (prover or verifier) message. Every possible option is represented by a different child; no two children represent the same option. The edge between a parent and a child actually represents the corresponding move in the IOP. The root will be the empty transcript, and (only) leaves will be (all possible) full transcripts

We associate with each node h in the IOP tree of π a value $f(h) \in [0, 1]$, which upper bounds the probability that the verifier will accept in an execution in which h is a prefix of the transcript. Formally:

$$f(h) = \begin{cases} 0 & \text{if } h \text{ represents a full transcript for which } \mathcal{V} \text{ rejects} \\ 1 & \text{if } h \text{ represents a full transcript for which } \mathcal{V} \text{ accepts} \\ \max_{\tilde{h} \in \text{children}(h)} (f(\tilde{h})) & \text{if } h \text{ isn't a leaf and the prover sends the next message} \\ \mathbb{E}_{\tilde{h} \leftarrow \text{children}(h)} (f(\tilde{h})) & \text{if } h \text{ isn't a leaf and the verifier sends the next message} \end{cases}$$

To see why $f(h)$ upper bounds the acceptance probability in the execution, notice that if h is a leaf, then it represents a full transcript, so the verifier accepts either with probability 1 (then 1 is an upper bound on the acceptance chance) or rejects with probability 1 (then 0 is an upper bound on the acceptance chance). If the next step is a prover message, then h 's children represent all possible prover messages, and so the maximum over all of their values bounds $f(h)$. If the next step is a verifier message then, since the verifier chooses the next message randomly, then h 's value is at most the expectation over the upper bounds on the values of h 's children. (Since this is a public-coin IOP, the randomness is uniformly distributed, and the expectation will be the average of the upper bounds of the children.) Note that this does not mean that in the next step (of $\mathcal{V}$) the bound cannot increase; it might be that $\mathcal{V}$ choose randomness that increases the acceptance probability. The values of $f(h)$ for internal nodes can be formally proven by induction.

Next, we introduce notation associated with executions in the new IOP π' . In the following, denote $y^r := \{y_1^r, \dots, y_m^r\}$ and $z^r := \{z_1^r, \dots, z_m^r\}$. We will consider y_0^r, y^r, i_{r-1} as three different messages and z_0^r, z^r as two different messages, i.e. each choice of one of them is represented by a different edge in the IOP tree of π' . The *segment* r is defined as $(y_0^r, y^r, z_0^r, z^r, i_r)$. The segment is almost exactly one round at π' (which corresponds to two merged rounds in π), except for the fact that in π' we send i_r in round $r + 1$. Since in the first round there is no i_0 to send, we will have $\frac{t}{4}$ segments, and one last verifier message, which we will denote by $y_0^{\frac{t}{4}+1}$.

Next, we define the *projection* to the tree of π of nodes that are the first/last in a segment. This is done in the following way:

- The projection of the root in the tree of π' is the root in the tree of π .
- For a node $h = (h', y_0^r)$, we define $\pi(h) = (\pi(h'), y_0^r)$.
- For a node $h = (h', y_0^r, y^r, z_0^r, z^r, i_r)$, we define $\pi(h) = (\pi(h'), y_0^r, z_0^r, y_{i_r}^r, z_{i_r}^r)$.

Such nodes will be called *projectable*. They represent points in the execution where the execution of π corresponds to the execution of π' , and in which we can estimate the acceptance probability in π' (using the acceptance probability in π). Intuitively, their values will be the same as those of their projections in the tree of π .

Formally, for a projectable node h , we define $e(h) := f(\pi(h))$, which intuitively is the upper bound on the accepting probability at that given state (h) in the original IOP. Indeed, at projectable nodes the two IOP executions arrived at the same node.

Step 2: Bound the probability of choosing y^r that will allow $\mathcal{P}'$ to improve his acceptance chances too much.

Intuitively, the next lemma bounds the probability the verifier chooses y^r for which the prover can choose messages (z_0^r, z^r) such that there will be “too many” options (specifically, more than $\frac{m}{2t}$ options for i_r) for which the acceptance probability increases by a multiplicative factor of $12t$.

Lemma 2. Let $r \leq \frac{t}{4}$, h be a node that represents a transcript that ends with some y_0^r , and $\delta = e(h)$. For some valid y^r denote $C(y^r)$ as the event in which

$$\exists z_0^r : |\{i | \exists z_i^r : f(\pi(h), z_0^r, y_i^r, z_i^r) > 12t\delta\}| > \frac{m}{2t}$$

Then for a random y^r we get $\Pr[C(y^r)] < 2^{-s}$, where the probability is over the choice of y^r .

Proof: We will prove the lemma in three stages:

1. We prove that $\Pr_{y_{i_r}^r} [\exists z_{i_r}^r : f(\pi(h), z_0^r, y_{i_r}^r, z_{i_r}^r) > 12t\delta \mid z_0^r, i_r] \leq \frac{1}{12t}$
2. We prove that $\Pr_{y^r} [|\{i_r \mid \exists z_{i_r}^r : f(\pi(h), z_0^r, y_{i_r}^r, z_{i_r}^r) > 12t\delta\}| > \frac{m}{2t} \mid z_0^r] \leq 2^{-m/2t}$
3. We prove that $\Pr_{y^r} [|\{i_r \mid \exists z_{i_r}^r : f(\pi(h), z_0^r, y_{i_r}^r, z_{i_r}^r) > 12t\delta\}| > \frac{m}{2t}] = \Pr[C(y^r)] \leq 2^{-s}$ (which proves the lemma).

Step 1: First, we show that:

$$\Pr_{y_{i_r}^r} [\exists z_{i_r}^r : f(\pi(h), z_0^r, y_{i_r}^r, z_{i_r}^r) > 12t \mid z_0^r, i_r] \leq \frac{1}{12t}$$

for any z_0^r, i_r .

From the definition of $f(\cdot)$, we get $f(\pi(h), z_0^r) \leq f(\pi(h)) = e(h) = \delta$ for any z_0^r , since z_0^r is a prover message. Therefore, $\forall z_0^r \forall i_r : \delta \geq f(\pi(h), z_0^r) = \mathbb{E}_{y_{i_r}^r} [f(\pi(h), z_0^r, y_{i_r}^r)]$ (the rightmost equality holds since the next message is $y_{i_r}^r$, which is a verifier message. Note that since we conditioned on i_r — which is chosen uniformly at random and independently of all other values — $y_{i_r}^r$ is distributed as the corresponding message in π). Then, by Markov's inequality, we get

$$\forall z_0^r \forall i_r : \Pr_{y_{i_r}^r} [f(\pi(h), z_0^r, y_{i_r}^r) > 12t\delta] \leq \frac{\mathbb{E}_{y_{i_r}^r} [f(\pi(h), z_0^r, y_{i_r}^r)]}{12t\delta} \leq \frac{\delta}{12t\delta} = \frac{1}{12t}.$$

Finally, since $z_{i_r}^r$ is a prover message, we get $\forall z_0^r \forall y_{i_r}^r \forall z_{i_r}^r : f(\pi(h), z_0^r, y_{i_r}^r, z_{i_r}^r) \leq f(\pi(h), z_0^r, y_{i_r}^r)$, and so:

$$\forall z_0^r \forall i_r \forall z_{i_r}^r : \Pr_{y_{i_r}^r} [f(\pi(h), z_0^r, y_{i_r}^r, z_{i_r}^r) > 12t\delta] \leq \Pr_{y_{i_r}^r} [f(\pi(h), z_0^r, y_{i_r}^r) > 12t\delta] \leq \frac{1}{12t}$$

This concludes stage 1.

Step 2: Now, we will prove that

$$\Pr_{y^r} [|\{i_r \mid \exists z_{i_r}^r : f(\pi(h), z_0^r, y_{i_r}^r, z_{i_r}^r) > 12t\delta\}| > \frac{m}{2t} \mid z_0^r] \leq 2^{-m/2t}$$

Intuitively, in this step we bound the probability of the event that we are interested in, but for a specific z_0^r , and without conditioning on i_r . The probability is over the choice of y^r , since z_0^r is fixed.

For z_0^r , define $B(z_0^r, y^r)$ as the event in which

$$|\{i_r \mid \exists z_{i_r}^r : f(\pi(h), z_0^r, y_{i_r}^r, z_{i_r}^r) > 12t\delta\}| > \frac{m}{2t}.$$

Then

$$\Pr[B(z_0^r, y^r)]_{y^r} \leq \binom{m}{m/2t} \cdot \left(\frac{1}{12t}\right)^{\frac{m}{2t}} \leq \left(\frac{em}{m/2t}\right)^{\frac{m}{2t}} \cdot \left(\frac{1}{12t}\right)^{\frac{m}{2t}} = (2te)^{\frac{m}{2t}} \cdot \left(\frac{1}{12t}\right)^{\frac{m}{2t}} = \left(\frac{e}{6}\right)^{\frac{m}{2t}} < \left(\frac{1}{2}\right)^{\frac{m}{2t}} = 2^{-\frac{m}{2t}}$$

We now explain the transitions (from left to right): in the left-most inequality, we choose $m/2t$ indices i out of m , and for each the probability that f 's value is more than $12t\delta$ is no more than $\frac{1}{12t}$ (over the choice of y^r). The next inequality follows from a known bound on the binomial coefficients:

$$\forall 1 \leq k \leq n : \binom{n}{k} \leq \left(\frac{ne}{k}\right)^k.$$

This concludes step 2.

Step 3: Now,

$$\Pr[C(y^r)] = \Pr_{y^r} [\exists z_0^r : B(z_0^r, y^r)] < |Z_0^r| \cdot 2^{-m/2t} \leq 2^{s-m/2t} = 2^{-s}$$

When $|Z_0^r|$ is the size of the domain from which z_0^r is chosen. We now explain the transitions (from left to right): the first inequality holds because for each z_0^r , $B(z_0^r, y^r) < 2^{-m/2t}$. The second follows by $|Z_0^r| \leq 2^{-s}$ (by s 's definition), and the rightmost equality holds because $m = 4st$. ■

Stage 3: Define lucky nodes and bound their fraction.

In the next stage of the proof, we will consider some nodes that make the f -value significantly larger, and show that such nodes are reached with very small probability.

Definition 6. We say that a node in the IOP tree of π' is *lucky* if (for a suitable history h for each case):

- Type 1: it is of the form (h, y_0^r) and $e(h, y_0^r) > 2te(h)$
- Type 2: it is of the form (h, y^r) and $\exists z_0^r : |\{i \mid \exists z_i^r : f(\pi(h), z_0^r, y_i^r, z_i^r) > 12te(h)\}| > \frac{m}{2t}$
- Type 3: it is of the form $(h, y^r, z_0^r, z^r, i_r)$ and $e(h, y^r, z_0^r, z^r, i_r) > 12te(h)$

We now bound the fraction of lucky nodes, i.e. at a given state, the fraction of the lucky nodes among all of its children. In the following, we refer to a node representing a transcript that ends with α as an “ α -node”.

Lemma 3. The fraction of lucky nodes is bounded as follows:

- Type 1: An i_r -node h has no more than a fraction of $\frac{1}{2t}$ lucky children. (Notice that all such children will be of [type 1](#).)
- Type 2: A y_0^r -node h has less than a fraction of 2^{-s} lucky children. (Notice that all such children will be of [type 2](#).)
- Type 3: For a z^r -node (h, y^r, z_0^r, z^r) , if (h, y^r) is not lucky, then the node (h, y^r, z_0^r, z^r) has no more than a fraction of $\frac{1}{2t}$ lucky children. (Notice that all such children will be of [type 3](#).)

Proof: We first bound the fraction of type 1 lucky nodes. Consider an i_r -node h . We bound the fraction of lucky children of h using the definitions of e and f and Markov's inequality as follows:

$$\begin{aligned} e(h) &= f(\pi(h)) = \mathbb{E}_{y_0^r}[(f(\pi(h), y_0^r))] = \mathbb{E}_{y_0^r}[f(\pi(h, y_0^r))] = \mathbb{E}_{y_0^r}[e(h, y_0^r)] \\ \Rightarrow \Pr[e(h, y_0^r) > 2te(h)] &\leq \frac{\mathbb{E}_{y_0^r}[e(h, y_0^r)]}{2te(h)} = \frac{e(h)}{2te(h)} = \frac{1}{2t}. \end{aligned}$$

Next, let h be a y_0^r -node. Then the fraction of lucky type-2 children of h is at most 2^{-s} by lemma 2. Indeed, since the verifier chooses y^r uniformly at random, the probability of sampling y^r that will cause the next node to be lucky is equal to the fraction of such y^r 's among options. By lemma 2 it is no more than 2^{-s} .

Finally, consider a z^r -node $hy^r z_0^r z^r$, and we bound the fraction of type-3 lucky children it has when hy^r is not lucky. Since hy^r is not lucky, then by definition of lucky [type 2](#) nodes, for all choices of z_0^r, z^r (in particular, the choice in the given z^r -node) the number of i 's for which $e(hy^r z_0^r z^r i_r) > 12te(h)$ (i.e., which will be a lucky [type 3](#) node) is not greater than $\frac{m}{2t}$. Since the z^r node has m children, then the fraction of lucky children is less than $\frac{m/(2t)}{m} = \frac{1}{2t}$. ■

Step 4: Transcript without lucky nodes - the verifier reject.

Now, let us look at a full transcript h . If h has no lucky node in it, then for every segment $(y_0^r, y^r, z_0^r, z^r, i_r)$ with the previous partial transcript h' we get

$$e(h', y_0^r, y^r, z_0^r, z^r, i_r) \leq 12t \cdot e(h', y_0^r) \leq 12t \cdot 2te(h') = 24t^2 e(h')$$

where the first two left-most inequalities hold because $(h', y_0^r, y^r, z_0^r, z^r, i_r)$ and (h', y_0^r) are not lucky nodes, and by the definitions of a lucky nodes of [type 3](#), [type 1](#), respectively.

Intuitively, this shows that for every segment, if we were back to π after that segment, then the probability of acceptance increases by at most $24t^2$. We have $\frac{t}{4}$ segments and one last layer of $y_0^{t/4+1}$, which is also not lucky and therefore does not increase the bound by more than $2t$. The value of e of the root is equal to the soundness error of π which is $\epsilon \leq (24t)^{-t}$. Overall, we get

$$e(h) \leq (24t^2)^{\frac{t}{4}} 2t \cdot \epsilon \leq (24t)^{\frac{t}{2}} 2t \cdot (24t)^{-t} < (24t)^{\frac{t}{2}} (24t)^{\frac{t}{2}} \cdot (24t)^{-t} = (24t)^t \cdot (24t)^{-t} = 1$$

Since h is a full transcript and therefore projectable, the acceptance probability in π at this point (which is bounded by $e(h) = f(\pi(h))$) is the same as the acceptance probability in π' , and also since it is a full transcript, the acceptance probability can only be 1 or 0. Since it is less than 1, it has to be 0 - so the verifier reject.

Step 5: Bounding the probability of the existence of a lucky node.

It therefore remains to bound the probability that some node in the tree is a lucky node. Given a full transcript h , order the nodes from root to leaf. So, since we have $\frac{t}{4}$ segments, each includes 5 nodes ($y_0^r, y^r, z_0^r, z^r, i_r$), and one last node for $y_0^{\frac{t}{4}+1}$, we will have $\frac{5t}{4} + 1$ nodes in a full transcript. Denote by L_j the event, over $\mathcal{V}$'s and $\mathcal{P}$'s randomness, in which the j -th node is lucky, and by M_j the event in which none of the first $j - 1$ nodes are lucky. Then:

$$\begin{aligned} \Pr[\text{there is a lucky node}] &= \sum_{j=1}^{\frac{5t}{4}+1} \Pr[\text{node } j \text{ is the first lucky node}] \\ &= \sum_{j=1}^{\frac{5t}{4}+1} \Pr[L_j | M_j] \cdot \Pr[M_j] \leq \sum_{j=1}^{\frac{5t}{4}+1} \Pr[L_j | M_j] \end{aligned}$$

where the probabilities are over the randomness of $\mathcal{V}'$ and $\mathcal{P}'$.

We now bound $\sum_{j=1}^{\frac{5t}{4}+1} \Pr[L_j | M_j]$. For this, we will use now lemma 3. Notice that while Lemma 3 bounds the fraction of lucky nodes, it also bounds the probability that a node is lucky. Indeed, lucky nodes are determined by verifier messages, which are uniformly random, and therefore the probability of a lucky node is exactly the fraction of lucky nodes (the fraction of lucky nodes among all possible options for node j is equal to the probability that node j is lucky). So, if node j is of type 1, then by lemma 3, $\Pr[L_j | M_j] \leq \Pr[L_j] \leq \frac{1}{2t}$. (Indeed, $\Pr[L_j | M_j]$ is the probability that j is the *first* lucky node, which is upper bounded by the probability that j is a lucky node, namely $\Pr[L_j]$.) If node j is of type 2, then by lemma 3 and the same reasoning, $\Pr[L_j | M_j] \leq \Pr[L_j] < 2^{-s}$. If it is of type 3, then by lemma 3 (and since M_j mean there is no previous lucky node, in particular the condition in the lemma is satisfied) $\Pr[L_j | M_j] \leq \frac{1}{2t}$. For all other cases, node j is not a node of a type that can be lucky, and so $\Pr[L_j | M_j] = 0$. We have $\frac{t}{4}$ segment, and so $\frac{t}{4}$ nodes of each type, except for type 1 which we have one more (the $y_0^{\frac{t}{4}+1}$,th node). So,

$$\Pr_h[\text{there is a lucky node}] \leq \frac{t}{4} \left(\frac{1}{2t} + 2^{-s} + \frac{1}{2t} \right) + \frac{1}{2t} = \frac{t}{4} \left(\frac{1}{t} + 2^{-s} \right) + \frac{1}{2t} = \frac{1}{4} + t2^{-s-2} + \frac{1}{2t} < \frac{1}{3}$$

The rightmost inequality holds under our assumptions on t, s in the theorem statement (namely, that $t2^{-s-2} + \frac{1}{2t} < \frac{1}{12}$, see also Remark 3.2 below).

step 6: Wrapping things up.

In general, we have shown that except with probability $\frac{1}{3}$, h does not have lucky nodes (*step 5*), and that if h has no lucky nodes, then $\mathcal{V}'$ rejects (*step 4*). Therefore, we get

$$\Pr[(\mathcal{P}'^*(\mathbb{x}), \mathcal{V}(\mathbb{x})) = \text{accept}] \leq \Pr[\text{there is a lucky node}] < \frac{1}{3}$$

■

Remark. The soundness analysis (with soundness error $1/3$) was under the assumption that $t2^{-s-2} + \frac{1}{2t} < \frac{1}{12}$. We now justify this assumption. Let us say that we want $t2^{-s-2} < \frac{1}{24}$ and $\frac{1}{2t} < \frac{1}{24}$. The latter means $12 < t$, which is very reasonable since the transformation is most useful when the original IOP has a large number of rounds. For the former, we will get $\log(24t) - 2 < s$ which is also a reasonable assumption, since t is usually at most polylogarithmic in the input length, whereas s is at least n^ϵ for some constant $\epsilon > 0$.

3.3 HVZK preservation

In this part, we will prove that when there are no explicit messages of the prover (i.e. all prover's messages are oracles), reduction 2 preserves HVZK. Note that every IOP with implicit messages of the prover can be transformed into one without implicit messages by converting all implicit messages to oracles, and let the verifier query for all the bits in those messages if needed. So, the theorem we will prove is:

Theorem 3. Let $\pi = (\mathcal{P}, \mathcal{V})$ be a public coin IOP for the promise problem $(YES_{\mathcal{R}}, NO_{\mathcal{R}})$ (corresponding to a NP relation $\mathcal{R}$) with no explicit prover messages, t moves (divisible by 4), and the $Q - HVZK$ property. Let $\pi' = (\mathcal{P}', \mathcal{V}')$ be the result of applying construction 2 to π . Then π' has the $Q - HVZK$ property.

Proof:

We need to show that there exists a PPT sim' s.t. $\forall (\mathbb{x} = (x_{exp}, x_{imp}), \mathbb{w}) \in YES_{\mathcal{R}}$:

$$REAL_{\pi', \mathcal{V}'}(\mathbb{x}, \mathbb{w}) \equiv IDEAL_{\pi', sim', \mathcal{V}'}(\mathbb{x})$$

that is:

$$(View_{\mathcal{V}'}^{\mathcal{P}'}(\mathbb{x}, \mathbb{w}), Q(q_{\mathcal{V}'}))_{r_{\mathcal{P}'}, r_{\mathcal{V}'}} \equiv (View_{\mathcal{V}'}^{sim'}(\mathbb{x}), q_{sim'})_{r_{\mathcal{V}'}, r_{sim'}}$$

where Q is the same overhead function as in π ; $r_{\mathcal{P}'}, r_{\mathcal{V}'}, r_{sim'}$ are the random coins of $\mathcal{P}', \mathcal{V}'$ and sim' (respectively); $q_{\mathcal{V}'}$ denotes the number of queries that $\mathcal{V}'$ makes to all her oracles (i.e., x_{imp} and $\mathcal{P}'$'s oracle messages); and $q_{sim'}$ denotes the number of queries which sim' makes to x_{imp} .

From the Q -HVZK of π and the assumption, there exists a PPT sim s.t. $\forall (\mathbb{x} = (x_{exp}, x_{imp}), \mathbb{w}) \in YES_{\mathcal{R}}$:

$$REAL_{\pi, \mathcal{V}}(\mathbb{x}, \mathbb{w}) \equiv IDEAL_{\pi, sim, \mathcal{V}}(\mathbb{x})$$

We define sim' as follows:

sim' : Input: $x_{exp}, 1^{|x_{imp}|}$, and oracle access to x_{imp} .

1. Initiate an empty history $H = (h_1, \dots, h_{\frac{t}{2}+1})$.
 sim' will use H later, treating the h_i 's as the random strings of $\mathcal{V}$ while emulating sim . The h_i 's will correspond to the i_r 's, intuitively emulating sim using the same verifier randomness from the chosen split in the transformed IOP.

Communication Phase

2. For $r = 1, \dots, \frac{t}{4}$:
 - (a) Wait to receive $y_0^r, \{y_1^r, \dots, y_m^r\}, i_{r-1}$ from $\mathcal{V}'$.
 In the first round, no i_{r-1} is received.
 - (b) If this is not the first round, set $h_{2r-2} = y_{i_{r-1}}^{r-1}$.
 - (c) Set $h_{2r-1} = y_0^r$.
3. Wait to receive $i_{\frac{t}{4}}, y_0^{\frac{t}{4}+1}$ from $\mathcal{V}'$.
4. Set $h_{\frac{t}{2}} = y_{i_{\frac{t}{4}}}^{t/4}$ and $h_{\frac{t}{2}+1} = y_0^{\frac{t}{4}+1}$.

Query Phase

5. Emulate sim on input $(x_{exp}, 1^{|x_{imp}|})$, with h_i as the randomness received from $\mathcal{V}$ at round i . Answer $\mathcal{V}'$'s oracle query by forwarding them to sim and returning the same responses.

sim' is PPT. t is the number of moves in $(\mathcal{P}, \mathcal{V})$ and since $\mathcal{V}, \mathcal{P}$ are PPT, $t = poly(n)$, and every message y_i^r or y_0^r that $\mathcal{V}'$ sends is also of $poly(n)$ length, and therefore the simulation in the communication phase takes $poly(n)$ time. sim is also PPT, and so also the query phase takes $poly(n)$ time. Overall sim' is PPT.

Real and Simulated view are identically distributed. Let $(\mathbb{x} = (x_{exp}, x_{imp}), \mathbb{w}) \in YES_{\mathcal{R}}$. The view of $\mathcal{V}'$ includes:

- $\{i_r\}_{r=1}^{t/4}, \{(y_0^r, y_1^r, \dots, y_m^r)\}_{r=1}^{t/4}, y_0^{\frac{t}{4}+1}$ that is, $r_{\mathcal{V}'}$ (The randomness of $\mathcal{V}'$).
- $x_{exp}, 1^{|x_{imp}|}$. (The input of $\mathcal{V}'$).
- ω_{imp} - responses to oracle queries of $\mathcal{V}'$ to x_{imp} .
- $\{\omega_r\}_{r=1}^{t/4}$, when ω_r are the responses to oracle queries of $\mathcal{V}'$ to a prover oracle sent in round r .

First, $x_{exp}, 1^{|x_{imp}|}$ are the input and so are the same both at real and ideal worlds. We condition on the input from now on. Since $\mathcal{V}'$ is non-adaptive, she does not make any query during the communication phase, and π' is public coin (because π is), her messages in this phase are independent of the prover's messages. So, $\{i_r\}_{r=1}^{t/4}, \{(y_0^r, y_1^r, \dots, y_m^r)\}_{r=1}^{t/4}$ and $y_0^{\frac{t}{4}+1}$ will also distributed the same, and we can condition on them too.

$\mathcal{V}'$ is honest, so she will only query according to what $\mathcal{V}$ would query when the randomness of the communication phase is the one that corresponding to the chosen $\{i_r\}_{r=1}^{t/4}$ set. In particular, she will not query a prover message that does not match the appropriate i_r . In the real world (when she is interacting with $\mathcal{P}'$) she will get the same answers $\mathcal{V}$ would get, and so from the Q -HVZK of π , these answers are distributed identically to the answers sim would have given. Since sim' uses sim (with the same verifier randomness we conditioned on above), these answers are identically distributed to the ones given by sim' in the ideal world. That is, $\omega_{imp}, \{\omega_r\}_{r=1}^{t/4}$ are also distribute the same. We get the following:

$$View_{\mathcal{V}'}^{\mathcal{P}'}(\mathbb{x}, \mathbb{w})_{r_{\mathcal{P}'}, r_{\mathcal{V}'}} \equiv View_{\mathcal{V}'}^{sim'}(\mathbb{x})_{r_{\mathcal{V}'}, r_{sim'}}$$

Also, since honest $\mathcal{V}'$ only makes queries when $\mathcal{V}$ does, sim' only makes queries when sim does, so from Q -HVZK of π :

$$Q(q_{\mathcal{V}'}) \equiv Q(q_{\mathcal{V}}) \equiv q_{sim} \equiv q_{sim'}$$

and overall:

$$REAL_{\pi', \mathcal{V}'}(\mathbb{x}, \mathbb{w}) \equiv IDEAL_{\pi', sim', \mathcal{V}'}(\mathbb{x})$$

■

Remark. The proof strategy given above does not work when there are explicit messages from the prover. Adjusting the proof to this case requires a more complicated argument in which sim' runs m copies of sim' , and rewinds one round once the “correct” y_i^r is determined. This might increase the query complexity of sim' , if sim' makes queries before the query phase. More importantly, in this case the transformation might actually violate HVZK, since the verifier is exposed to explicit information (messages) from multiple different parallel executions.

4 Summery

Contributions and Conclusions. In this project, we looked at two transformations intended to decrease the rounds complexity of an IOP and referred to the transformations' effect on the ZK property of the ZK-IOP it operates on. We focused on public coin IOPs with perfect completeness. We started with the reduction represented at [ABCY22] and proved it is ZK-preserving. After that, since the first reduction is relevant only for specific cases, we looked at the reduction at [BM88]. We first converted the reduction to the IOP model, since it was originally planned for IPs. We proved that this reduction is valid also in the IOP model and then demonstrated that it preserves ZK for an honest verifier (under a reasonable assumption).

To sum up, the contributions are:

1. ZK preservation proof for reduction 1.
2. Adaption of the reduction at [BM88] to IOP model which included its time analysis and completeness and soundness proof.
3. HVZK preservation of reduction 2.

ZK is a property affected even by tiny changes, and preservation is not trivial.

Future Research Directions. Future directions of this work might be about the reduction 2 HVZK-preservation when there are explicit messages of the prover, and also general ZK-preservation (and not only HVZK), which seems not to be happening.

A more general and significant direction is to find a (existing or new) reduction that reduce the round complexity and is also ZK preserving.

References

- [ABCY22] Gal Arnon, Amey Bhangale, Alessandro Chiesa, and Eylon Yogev. A toolbox for barriers on interactive oracle proofs. pages 25–26, 2022.
- [ARZR25] Noor Athamnah, Noga Ron-Zewi, and Ron D. Rothblum. Linear prover iops in log star rounds. 2025.
- [BCL21] Jonathan Bootle, Alessandro Chiesa, and Siqi Liu. Zero-knowledge IOPs with linear-time prover and polylogarithmic-time verifier. 2021.
- [BM88] Laszlo Babai and Shlomo Moran. Arthur-merlin games: A randomized proof system, and a hierarchy of complexity classes. Journal of computer and system sciences 36, pages 267–271, 1988.
- [BSCS16] Eli Ben-Sasson, Alessandro Chiesa, and Nicholas Spooner. Interactive oracle proofs. 2016.
- [RRR17] Omer Reingold, Guy N. Rothblum, and Ron D. Rothblum. Personal communication, 2017.
- [RW24] Noga Ron-Zewi and Mor Weiss. Zero-Knowledge IOPs Approaching Witness Length. pages 15–18, 2024.
- [VW26] Himanshu Vashishth and Mor Weiss. Composition theorems for zero-knowledge IOPs. page 16, 2026.